

What is meant by a 'sustainable diet'?



What is meant by an ‘environmentally sustainable diet’?

‘Sustainable Diets are those diets with low environmental impacts which contribute to food and nutrition security and to healthy life for present and future generations. Sustainable diets are protective and respectful of biodiversity and ecosystems, culturally acceptable, accessible, economically fair and affordable; **nutritionally adequate**, safe and **healthy** while optimizing natural and human resources.’

FAO, 2010³¹

How do we assess sustainability in food?

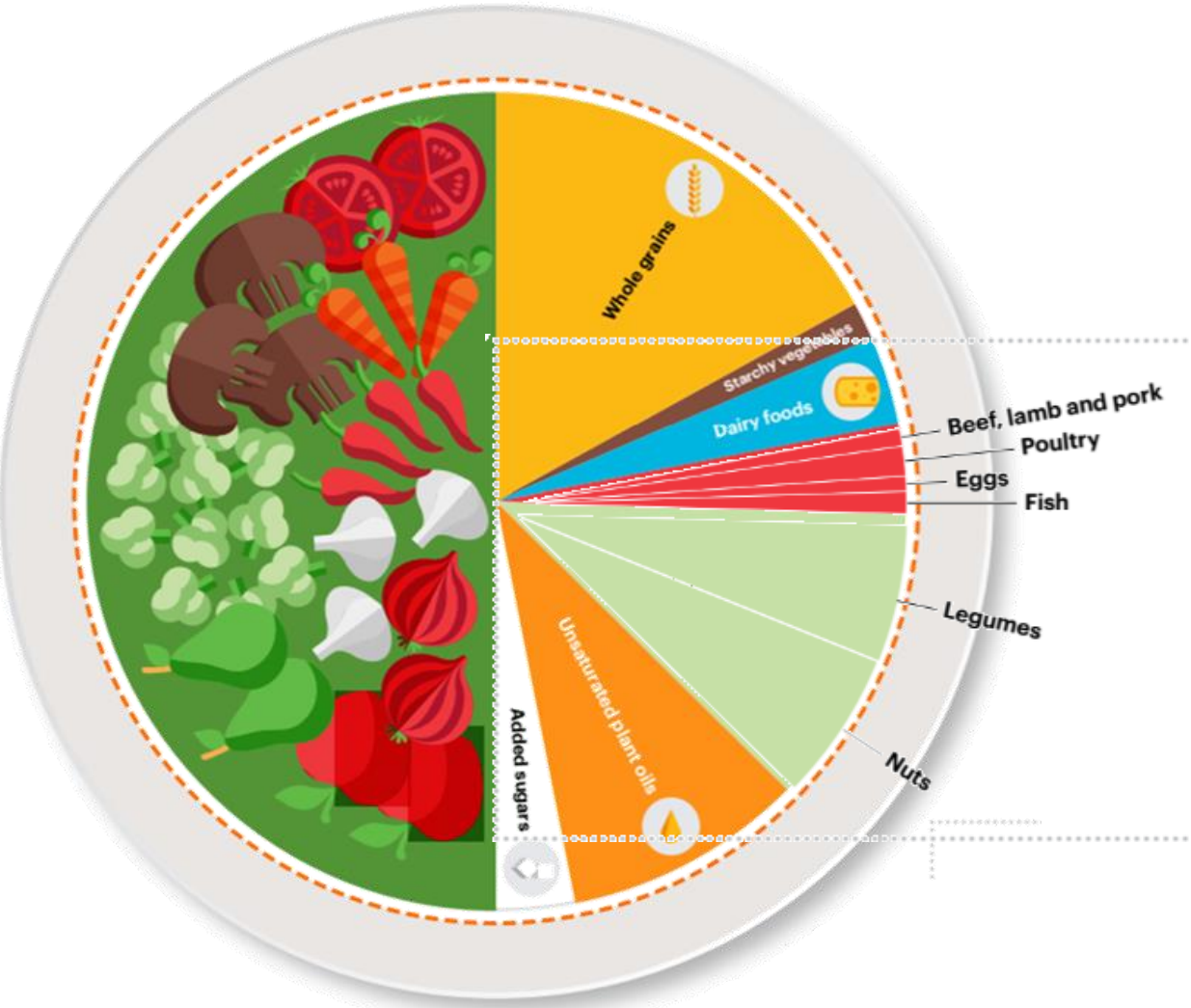
The *average* quantity of GHGs emitted and land & water use to produce 100g of protein from animal and plant foods.

These are *average* values based on a combination of data sets from around the world therefore **there will be significant variability depending on country of origin and production practices.**

Other important measure include water pollution, soil degradation, impact on biodiversity & ecosystems, nitrogen and phosphorus output (from pesticide use) etc.

	GHGe as kg CO ₂ eq per 100g of protein	Land use m ² per 100g of protein	Stress weighted water use 1000s litres per 100g of protein
Nuts	0.3	7.9	140.8
Peas	0.4	3.4	12.6
Beans	0.8	7.3	10.5
Peanuts	1.2	3.5	23.6
Tofu	2	2.2	3.2
Eggs	4.2	5.7	16.2
Poultry meat	5.7	7.1	8.2
Fish (farmed)	6	3.7	18.2
Pig meat	7.6	11	41.3
Cheese	11	40	81.9
Dairy cattle	17	22	60.7
Shellfish (farmed)	18	2	86.2
Sheep meat	20	185	70.9
Beef meat	50	164	17.4

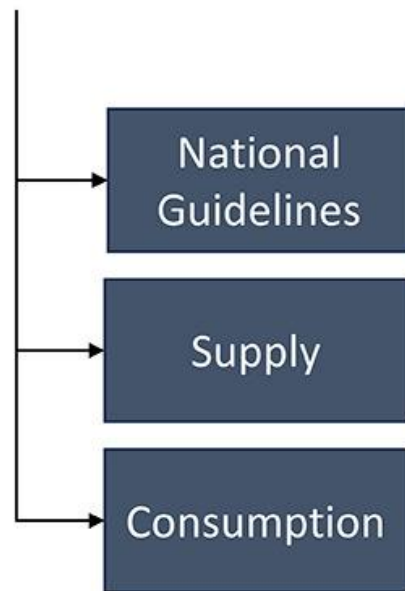
EAT Lancet recommendations: Planetary Health Diet



	Macronutrient intake grams per day (possible range)	Caloric intake kcal per day
Whole grains Rice, wheat, corn and other	232	811
Tubers or starchy vegetables Potatoes and cassava	50 (0-100)	39
Vegetables All vegetables	300 (200-600)	78
Fruits All fruits	200 (100-300)	126
Dairy foods Whole milk or equivalents	250 (0-500)	153
Protein sources		
Beef, lamb and pork	14 (0-28)	30
Chicken and other poultry	29 (0-58)	62
Eggs	13 (0-25)	19
Fish	28 (0-100)	40
Legumes	75 (0-100)	284
Nuts	50 (0-75)	291
Added fats		
Unsaturated oils	40 (20-80)	354
Saturated oils	11.8 (0-11.8)	96
Added sugars		
All sugars	31 (0-31)	120

Do Malaysian diets align with Planetary Health Diet recommendations?

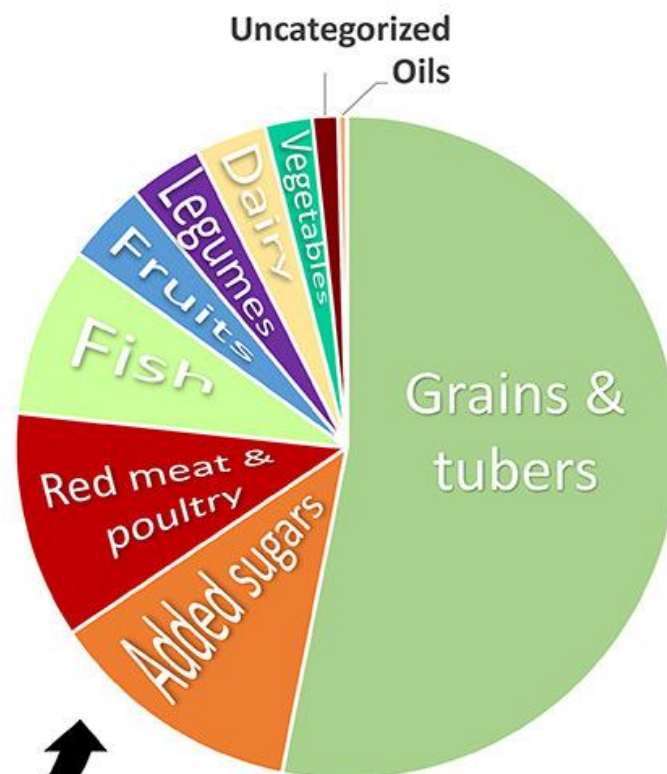
MALAYSIAN DIET DATA



Malaysian Dietary Guidelines 2020

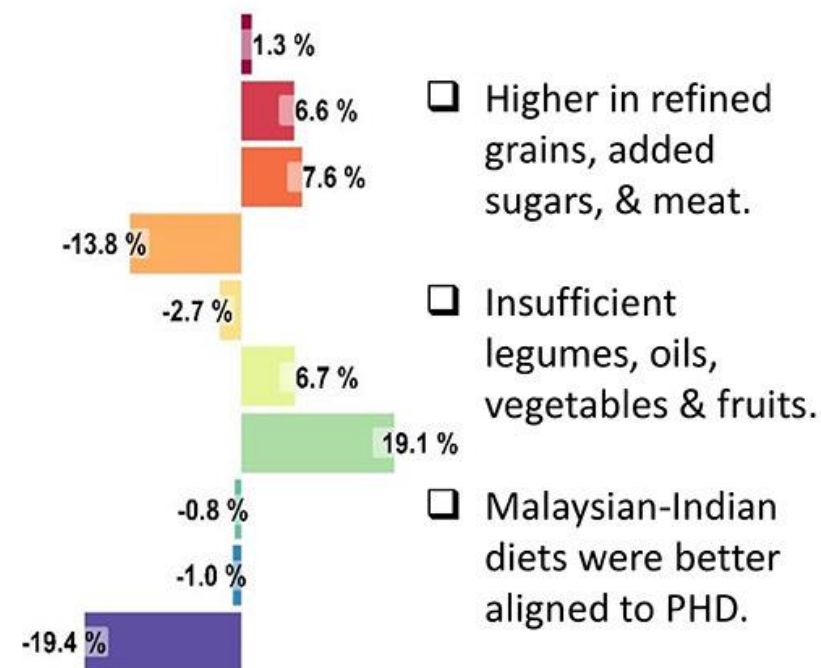
FAO Food Balance Sheet 2020

Malaysian Adult Nutrition Survey 2014



Distribution of food groups
(as a percentage of daily calories, %)

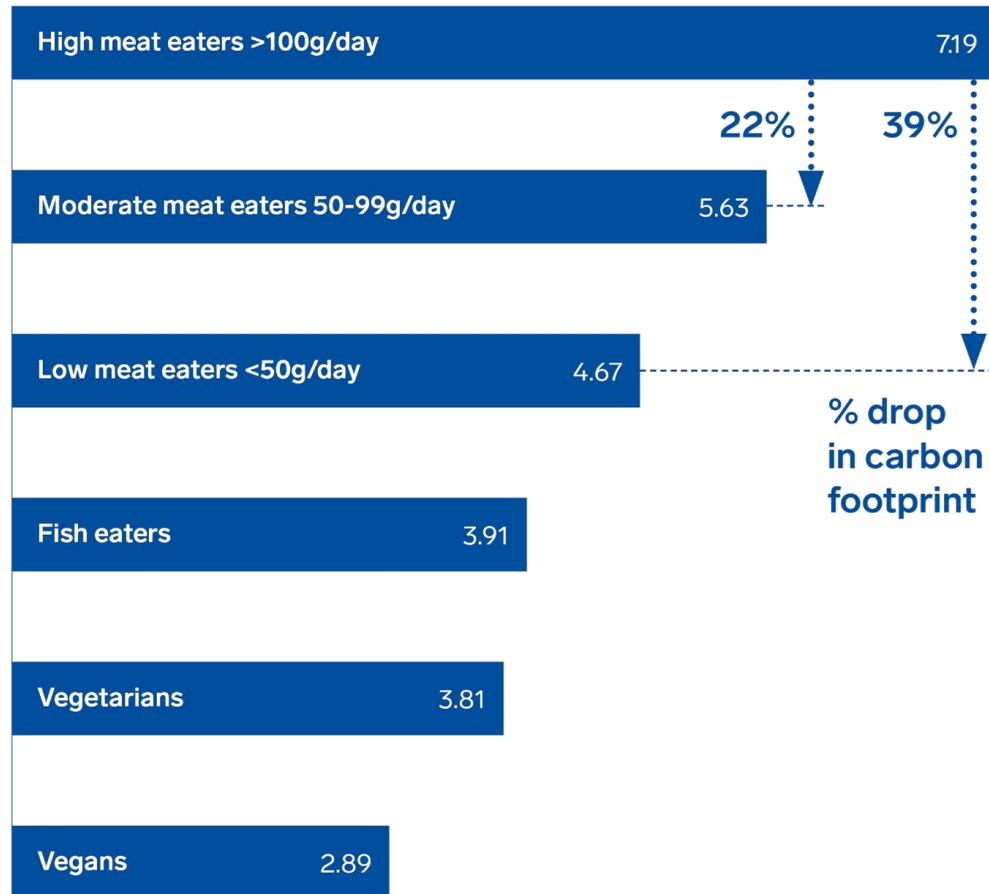
Malaysian Diet vs Planetary Health Diet



Conclusions

- ✓ It is possible for Malaysians to adopt a predominantly plant-based diet with lower health and environmental costs.
 - ✓ The Planetary Health Diet guidelines, with appropriate adaptation, can be used to transform Malaysia's national agro-food systems and dietary guidance by providing a consistent framework to assess, address, and monitor the environmental and human health impacts of our food systems.
- (Rajaram et al. 2024)

Veganism and Vegetarianism



Diet-related GHG Emission in Malaysia

GHG EMISSIONS

